MATERNAL LABOR FORCE PARTICIPATION: U.S. VS MEXICO

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ABSTRACT

In this project, I examine **how family structure and child characteristics shape mothers' likelihood of working in the United States and Mexico** using 2020 IPUMS International data.

I estimated **logistic regression models** to evaluate how the number of children, the age of the eldest child, marital status, education, and age influence employment outcomes. I also included an interaction between being unmarried and having an eldest child aged 13 or older to test whether older children provide additional support for mothers.

Overall, there is clear evidence that l**arger families reduce the probability of working**, while **higher education and older children increase it**. However, the strength of these relationships differs between countries.

LITERATURE REVIEW

Fertility and Maternal Labor Supply:

- Large international evidence shows that **additional children reduce mothers' labor force participation**, with this pattern holding across the past two centuries and across multiple countries (Aaronson et al., 2017).

Child Age and Employment:

- Mothers' labor force **participation increases as children get older**, reflecting declining childcare demands as children reach school age and adolescence (Bureau of Labor Statistics, 2025).

Household Structure and Older Children:

- Research on household specialization shows that f**amily composition influences labor decisions**; older children may contribute to household support in ways that can ease mothers' return to work (Lundberg & Rose, 2002).

Education, Labor Markets, and Mexico:

- Structural changes in Mexico, such as **rising education levels**, falling fertility, and shifts in labor demand, have contributed to long-run increases in women's employment (Bhalotra & Fernández, 2024).
- Additional evidence shows that fertility may shape maternal employment differently in contexts with large informal sectors, such as Mexico (Schmieder, 2021).

OBJECTIVE

To examine how fertility, child age, marital status, and education influence the likelihood that women return to work in the United States and Mexico, using harmonized microdata from IPUMS International.

HYPOTHESIS

Women with more children will have a lower likelihood of being employed, even after accounting for child age, marital status, and education.

**This hypothesis applies to both the United States and Mexico.*

Cross-country causal evidence from Latin America shows that the negative effect of fertility on women's employment in Mexico closely resembles findings for the United States, indicating similar underlying behavioral mechanisms across contexts (Cruces & Galiani, 2007; Tortarolo, 2014).

METHODOLOGY

Women ages 18–50
from IPUMS 2020
United States & Mexico

empstat to working → 1 = employed, 0 = not working
nchild → number of own children
elch to eldest13 → 1 if eldest child ≥13, 0 otherwise
marstat to married → 1 if married/in union, 0 otherwise
eldest13_notmarried → interaction term
edattain to highedu → higher vs. lower education
age → respondent age

Logistic Regression
logit working X if country == 484 or 840

a note on interpretation

A logistic regression **does not estimate changes in probability directly**. Instead, each coefficient represents the change in the log-odds of working, holding all other variables constant. A negative coefficient means the odds of working decrease, while a positive coefficient means the odds increase.

A coefficient like −0.11 does NOT mean
“probability drops by 11%.”
It means the log-odds decrease by 0.11.

Mexico

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. logit working nchild eldest13 married eldest13_notmarried highedu age if country == 484, vce(robust)

Iteration 0:  Log pseudolikelihood = -1661350.2
Iteration 1:  Log pseudolikelihood = -1488906.9
Iteration 2:  Log pseudolikelihood = -1487872.4
Iteration 3:  Log pseudolikelihood = -1487871.8
Iteration 4:  Log pseudolikelihood = -1487871.8

Logistic regression                                     Number of obs = 2,499,285
                                                        Wald chi2(6)  = 285920.56
                                                        Prob > chi2   =    0.0000
Log pseudolikelihood = -1487871.8                       Pseudo R2     =    0.1044
```

working	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
nchild	-.1099565	.0012957	-84.87	0.000	-.1124959	-.107417
eldest13	.2434697	.0043976	55.36	0.000	.2348504	.2520889
married	-1.466183	.005018	-292.18	0.000	-1.476019	-1.456348
eldest13_notmarried	-.0338481	.0072036	-4.70	0.000	-.0479668	-.0197294
highedu	.9695812	.003134	309.37	0.000	.9634386	.9757238
age	.0289888	.0002255	128.55	0.000	.0285468	.0294308
_cons	-.5030256	.0084517	-59.52	0.000	-.5195907	-.4864606

Mexico

Predictor	Coefficient	Interpretation
nchild	−0.110	Each additional child lowers the probability of working, reflecting heavier childcare demands.
eldest13	0.243	Having an eldest child aged 13+ increases the probability of working because older children require less supervision.
married	−1.466	Married women are substantially less likely to work compared to unmarried women, suggesting stronger household specialization or cultural norms around employment.
eldest13_notmarried	−0.034	For unmarried women, having an eldest child age 13+ slightly lowers employment likelihood. This may reflect added responsibilities without partner support.
highedu	0.97	Higher education sharply increases employment probability. This is one of the strongest predictors in the model.
age	0.029	Employment likelihood rises as women age, which is consistent with gaining experience, stability, and fewer very young children.

United States

```
. logit working nchild eldest13 married eldest13_notmarried highedu age if country == 840, vce(robust)

Iteration 0:  Log pseudolikelihood =  -906963.3
Iteration 1:  Log pseudolikelihood = -870791.36
Iteration 2:  Log pseudolikelihood = -870268.82
Iteration 3:  Log pseudolikelihood = -870268.72
Iteration 4:  Log pseudolikelihood = -870268.72

Logistic regression                                     Number of obs = 1,515,840
                                                        Wald chi2(6)  =  66447.33
                                                        Prob > chi2   =    0.0000
Log pseudolikelihood = -870268.72                       Pseudo R2     =    0.0405
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working	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
nchild	-.2354963	.0018346	-128.36	0.000	-.2390922	-.2319005
eldest13	.1567062	.0051733	30.29	0.000	.1465667	.1668457
married	-.291963	.0056401	-51.77	0.000	-.3030173	-.2809087
eldest13_notmarried	.0337568	.0085185	3.96	0.000	.0170608	.0504528
highedu	1.180319	.0064905	181.85	0.000	1.167598	1.19304
age	.0257295	.0003245	79.30	0.000	.0250936	.0263654
_cons	-.5157766	.0130739	-39.45	0.000	-.541401	-.4901523

United States

Predictor	Coefficient	Interpretation
nchild	−0.235	Each additional child lowers a woman's probability of working, and the effect is about twice as large as in Mexico.
eldest13	0.157	Having an eldest child age 13+ increases employment probability, but the effect is smaller than in Mexico.
married	−0.292	Married women in the U.S. are less likely to work than unmarried women, but the effect is far weaker than in Mexico.
eldest13_notmarried	0.034	Unmarried mothers with a 13+ child are slightly more likely to work. This points to possible support from older children compensating for limited household adults.
highedu	1.18	Higher education has a very strong positive effect on working; it is the strongest predictor in the U.S. model.
age	0.026	Employment likelihood increases with age.

Results Comparison

- In both countries, **having more children lowers the probability of working**, but the effect is much larger in the United States.
- **Older children increase employment likelihood** in both settings, although the size of the effect differs.
- **Education remains one of the strongest positive predictors of employment** everywhere, and being married reduces women's employment, especially in the United States.

- **The interaction term behaves differently: unmarried U.S. mothers with older children are more likely to work, while in Mexico they are less likely to work.**

CONCLUSION

- Fertility strongly **reduces** work participation.

- Older children ease mothers' return to work.

- Marital status plays a major role.
  - **Married women are less likely to work in both countries**. The interaction results suggest that unmarried mothers may benefit more from having an older child compared to married mothers.

- Education remains one of the strongest predictors.

- Age increases employment odds.

Cross-country comparison:
Mexico shows stronger negative effects of fertility and weaker positive effects of older children relative to the U.S., suggesting structural differences in childcare access and labor markets.

Overall Conclusion:

Both demographic and household factors shape maternal labor force participation, but the size and direction of these effects vary meaningfully between Mexico and the United States.

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THANK YOU!